

ABHIMANYU MANDAL

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PROFILE

Computational biologist with research experience in genomics, transcriptomics, whole-exome sequencing (WES), and multi-omics data analysis. Experienced in developing reproducible bioinformatics pipelines using R, Python, and Linux for the analysis of large-scale biological datasets. Passionate about integrating genomics, transcriptomics, multi-omics, and AI/ML approaches to advance biomarker discovery, precision medicine, and drug discovery.

TECHNICAL SKILLS

Genomics & Variant Analysis: Whole-exome sequencing (WES), variant calling, and annotation, GATK, IGV, ClinVar, gnomAD, OMIM, ACMG/AMP variant classification

Transcriptomics: Bulk RNA-seq, single-cell RNA-seq (Seurat), Differential gene expression, clustering, PCA, UMAP, t-SNE, trajectory inference, Pathway enrichment analysis

Programming: R (advanced), Python, Bash, SQL

Workflow & Infrastructure: Linux, HPC (SLURM/Torque), Docker, Nextflow, Git/GitHub

Visualization: R Shiny, ggplot2, Tableau

RESEARCH EXPERIENCE

Project Research Assistant, NDMC, IIT Bombay

Sep 2024 – Mar 2025

Computational epidemiology – Bayesian spatiotemporal modelling, R, HPC

- Built Bayesian spatiotemporal risk models across 70+ districts in Maharashtra and Gujarat, directly informing district-level TB risk stratification used by national public health stakeholders.
- Processed and analysed 10,000+ clinical and public health records in R, delivering fully reproducible pipeline outputs and structured technical reports under NDMC program timelines.
- Designed interactive dashboards and heatmaps that translated complex epidemiological findings for non-technical policy audiences — a skill directly transferable to R/Shiny visualisation of scRNA-seq results.

Bioinformatics Research Intern, Hu Lab, University of Western Ontario

May 2023 – Aug 2023

Cancer single-cell genomics – Seurat, scRNA-seq, differential expression, pathway analysis

- Processed, integrated, and analyzed over 40,000 single-cell transcriptomes from HER2+ and ER+ breast cancer models using Seurat, identifying lineage-specific biomarkers through differential expression and pathway enrichment analyses.
- Integrated single-cell transcriptomic signatures with public perturbation datasets to prioritize candidate therapeutic targets for downstream functional validation.
- Bridged computational and wet-lab research by translating scRNA-seq findings into experimentally testable biological hypotheses in close collaboration with bench scientists.

Biomedical Data Analyst Intern, Elucidata Corporation

Feb 2022 – Jul 2022

Multi-omics data infrastructure – R Shiny, QC pipelines, technical documentation

- Developed an R Shiny platform for quality assessment, visualization, and exploration of more than 1,000 multi-omics datasets, reducing manual QC time by approximately 30%.
- Performed systematic data quality checks and exploratory analysis, developing clear technical documentation that improved reproducibility across the data science team.

EDUCATION

Integrated Dual Degree (B.Tech + M.Tech), Pharmaceutical Engineering & Technology, IIT (BHU), Varanasi

2019 – 2024

CGPA: 8.74 / 10 | Specialisation: Genomics & Computational Biology

- Master's Thesis: Germline Variant Analysis in Indian OSCC-GB Patients (WES) — analysed 300+

WES samples, applied GATK/ClinVar/gnomAD annotation and ACMG/AMP classification to identify rare high-impact pathogenic variants linked to early onset and poor prognosis (manuscript under peer review).

SELECTED PROJECTS

Cancer Genomics – WES Variant Analysis

- Identified high-impact pathogenic variants in oncology WES datasets using multi-database annotation workflows; improved pipeline classification accuracy by integrating ClinVar, gnomAD, and multiple clinical prediction tools.

Gene Expression Analysis – Lymphoma Subtypes

- Conducted differential expression analysis across multiple lymphoma subtype microarray datasets, identifying consistent oncogenic expression signatures relevant to tumour progression.

Machine Learning for Neurotoxicity Prediction

- Developed a Support Vector Machine model using molecular descriptors from over 500 compounds to predict neurotoxicity, achieving 83% classification accuracy and demonstrating practical experience in machine learning for biological data.

Healthcare Claims Analytics (R & Tableau)

- Analyzed 15,000 health insurance claims using R and Tableau, building interactive dashboards for settlement, rejection, fraud, and hospital performance analysis, and generating actionable insights through EDA, KPI tracking, and business intelligence reporting.

PUBLICATIONS

- Mandal A., Khattri A. Predisposing Germline Mutations Associated with Early Onset and Poor Prognosis in Indian OSCC-GB Patients. (Under Review)
- Nag S., Mandal A., Joshi A., et al. Sialyltransferases and Neuraminidases: Potential Targets for Cancer Treatment. *Diseases*. 2022; 10(4):114.
- Nag S., Baidya ATK, Mandal A., et al. Deep learning tools for advancing drug discovery and development. *3 Biotech*. 2022; 12(5).

AWARDS & FELLOWSHIPS

- Project Research Assistant Fellowship – National Disease Modelling Consortium, IIT Bombay (2024)
- Junior Research Fellowship – IIT Bombay (2024)
- Mitacs Globalink Research Internship – University of Western Ontario, Canada (2023)
- Biomedical Data Analyst Fellowship – Elucidata Corporation (2022)

LEADERSHIP & MENTORSHIP

- Mentored 20+ undergraduates on research methodology, project planning, and graduate applications at IIT-BHU Research Cell (2022–2024), building the teaching and supervision skills valued in a multi-PI lab environment.
- Led student teams to curate 1,000+ research opportunities and coordinated alumni outreach, developing the organisational and communication skills directly applicable to collaborative, multidisciplinary research settings.